

3D error geometry: change vs. final norm

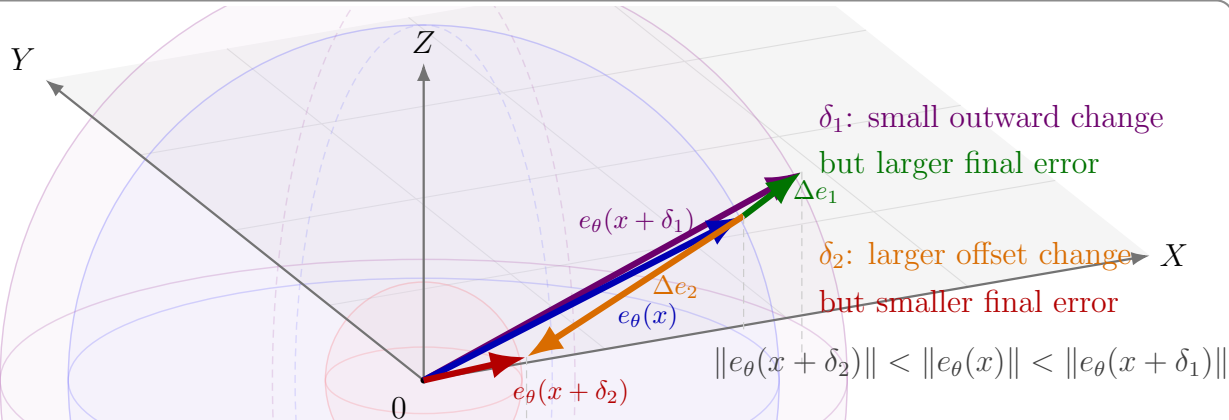
Two perturbations start from the same current error $e_\theta(x)$.

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|--|--------------------------|---|--------------|
|  | $e_\theta(x)$ |  | Δe_1 |
|  | $e_\theta(x + \delta_1)$ |  | Δe_2 |
|  | $e_\theta(x + \delta_2)$ | | |

$$e_0 = e_\theta(x) = \mathcal{F}_\theta(x) - \mathcal{G}(x)$$

$$e_i = e_\theta(x + \delta_i), \quad i = 1, 2$$

$$\Delta e_i = e_i - e_0$$



$\max_\delta \|\Delta e\|$ maximizes change size, not necessarily final error size.
 $\max_\delta \|e_\theta(x + \delta)\|$ is the final-error objective.